

Credit Card Approval Prediction

Project Conclusion Report

1. Project Summary

This project automates the credit card approval decision process for banks and financial institutions using machine learning. Historical applicant data from the Kaggle “Credit Card Approval Prediction” dataset (application_record.csv and credit_record.csv) was used to train and evaluate four classification algorithms: Logistic Regression, Decision Tree, Random Forest, and XGBoost. The best-performing model was integrated into a Flask web application that provides instant, real-time approval or rejection predictions based on applicant financial and demographic details.

2. Methodology Overview

- **Data Collection:** Merged 36,457 applicant records by combining demographic/financial data with a derived risk label, where an applicant was marked “risky” if they were ever 30+ days overdue on payments.
- **Exploratory Data Analysis:** Examined class imbalance (~88% good, ~12% risky), income and age distributions, and correlations between features, revealing that risk is not driven by any single linear factor.
- **Data Pre-processing:** Handled missing OCCUPATION_TYPE values, corrected a placeholder anomaly in DAYS_EMPLOYED, engineered AGE_YEARS and YEARS_EMPLOYED features, and applied one-hot encoding to categorical variables, expanding the feature set to 47 columns.
- **Model Building:** Trained four classification models using class-balancing techniques (class_weight='balanced' and scale_pos_weight) to address the significant class imbalance in the dataset.
- **Application Building:** Deployed the best model in a Flask web application with an interactive HTML form, enabling real-time eligibility checks.

3. Model Performance Comparison

All four models were evaluated on a held-out 20% test set (7,292 applicants) using accuracy, precision, recall, and F1-score, with particular emphasis on recall for the risky class, since failing to identify a high-risk applicant carries greater real-world cost than over-flagging a safe one.

Model	Accuracy	Precision (Risky)	Recall (Risky)	F1 (Risky)
Logistic Regression	0.567	0.136	0.499	0.213
Decision Tree	0.686	0.170	0.429	0.243
Random Forest	0.745	0.236	0.522	0.325

XGBoost	0.789	0.304	0.620	0.408
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XGBoost achieved the best performance across every metric, with an accuracy of 78.9%, a recall of 62.0% for risky applicants, and the highest F1-score (0.408) among all models tested. This confirms that gradient boosting is well-suited to capturing the complex, non-linear relationships present in credit risk data, consistent with the weak linear correlations observed during exploratory analysis.

4. Key Findings

- No individual numeric feature showed strong linear correlation with credit risk, indicating that risk arises from complex interactions between income, employment, demographics, and credit history rather than any single factor.
- The dataset exhibits significant class imbalance (~88% good vs. ~12% risky), which required class-weighting strategies during model training to avoid a model that simply predicts “approved” for everyone.
- Tree-based ensemble methods (Random Forest, XGBoost) substantially outperformed the linear Logistic Regression baseline, confirming the non-linear nature of credit risk patterns.
- Data quality issues, such as a placeholder value in the DAYS_EMPLOYED field for unemployed and retired applicants, required careful feature engineering to avoid misleading the model.

5. Application Outcome

The trained XGBoost model was serialized and integrated into a Flask web application, allowing a user to enter an applicant's financial and demographic profile through an HTML form and receive an instant prediction of “Approved (Low Risk)” or “Rejected (High Risk)”, along with a numeric risk score. This directly supports the project's stated use cases: automated screening by credit analysts, batch risk identification for compliance review, and self-service eligibility checks for prospective customers.

6. Limitations and Future Scope

- The dataset does not contain an explicit bank approval/rejection decision; applicant risk derived from repayment history was used as a proxy label, consistent with standard practice for this dataset.
- Precision for the risky class remains moderate (0.30), meaning a portion of safe applicants may still be flagged for manual review; further tuning or additional features could improve this.
- Future work includes deploying the model on a scalable cloud platform such as IBM Watson Machine Learning, enabling hosted, real-time predictions accessible beyond a local environment.
- Hyperparameter tuning and additional feature engineering (e.g., income-to-family-size ratio) could further improve model precision and recall balance.

7. Conclusion

This project successfully demonstrates an end-to-end machine learning pipeline for credit card approval prediction, from raw data ingestion through model deployment in a usable web application. Among the four algorithms evaluated, XGBoost provided the most reliable and balanced performance, making it the recommended model for production use. The resulting system offers a practical, automated alternative to manual credit application review, with the potential for further enhancement through cloud deployment and continued model refinement.